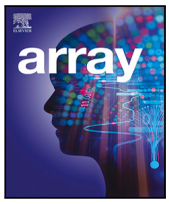
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Nonparametric SPC: Solving complex industrial monitoring tasks with low computational cost

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ABSTRACT

Quality assurance is essential for an organization's success today, but it is also a great challenge in the Big Data era. Statistical Process Control (SPC), focusing on control charts of monitoring variability, is a fundamental tool, though outliers, skewed information, and temporal dependence may influence the ones based on the process average. Nonparametric techniques stand out when it is needed to represent the process dynamism with robustness and derive statistical tests based on, e.g., the median instead of the mean. This work approaches the univariate Nonparametric Process Monitoring, such as Sign and Signed-Rank statistics, which are useful for run-type signaling, monitoring spread or joint (considering the known parameter of the specified tolerant interval, case K, or unknown parameter of in-control, case U). This study illustrates some mining industry monitoring tasks through nonparametric control charts, allowing the suggestion of a preventive maintenance plan for different scenarios. Due to the scarcity of implementations of nonparametric methods in the context of SPC, this study developed a new package for the R software, named `npSPC`, which shows extremely low computational cost. The package provides nonparametric control charts (alternative to the mean parametric charts of Shewhart, CuSum, and EWMA), enabling statistical monitoring in a wider range of problems under minimal assumptions (such as not the response variable being continuous or symmetric) with fast inferences.

1. Introduction

Quality assurance (QA) is a crucial element for the success of an organization. One of the fundamental tools in this context is Statistical Process Control (SPC), which encompasses a variety of techniques, with control charts being one of the most prominent. There are examples of applications of these charts to monitor processes in various areas, such as ecology [1], healthcare [2,3], environment [4,5], finance [6], and industry [7,8]. They are used for different types of data, such as counts [9,10], attributes [11], and rates/proportions [12,13].

There are different types of control charts regardless of the data-driven methodology (parametric, nonparametric, statistical, or machine learning method; for further information, see [14]). The most well-known/used of these are the Shewhart (e.g., $\overline{X}$ and Range/S charts), Cumulative Sum (CuSum), and Exponentially Weighted Moving Average (EWMA) types. Some charts are based on monitoring the average of a process sampled over time [15]. Nevertheless, the process summarization estimation may be influenced easily by outliers or temporal dependence [16] (though easily solved by changing the monitoring measure representation, for instance, from the mean to the median, which is less sensitive to process shifts). Additionally, in real-time data analysis without in-control specification, the parametric estimation computational cost can be high if updated across time, depending on the need for decision-making speed.

These three types of control charts (Shewhart, CuSum, EWMA) are constructed using parametric tests, meaning they assume a probability distribution (or asymptotic assumptions) for the data. For further details on their construction, see, for example, Montgomery [17], in which the parametric SPC tools are mostly mean-based. On many occasions, this characteristic is a strong disadvantage since data frequently may not follow a known distribution and present a small sample with unknown in-control bounds update in time; in such situations, it is more appropriate to use a nonparametric test, which relaxes assumptions about the data distribution. Based on the distribution-free type of test, median-based control charts can be constructed [18]. These charts may be more suitable in cases where the data distribution is unknown, does

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not follow a specific pattern, or in cases of limit estimation regardless of contamination.

In the Big Data era, Internet of Things (IoT) and real-time data are often expected to introduce a dynamic landscape characterized by asymmetric and non-normal data distributions [19]. This intricate fusion challenges traditional statistical assumptions and demands a high-speed computation toolbox for the SPC decision-making to monitor and establish tailored tolerance limits. As organizations navigate this complex terrain, precision with flexible SPC tools based on distribution-free hypothesis testing statistics may help define tolerance thresholds and become paramount for ensuring the reliability and effectiveness of data-driven decision-making processes. The distribution-free SPC charts are presented in the literature; nevertheless, they are not implemented in a package nor user-friendly, especially for non-programmer engineers.

To support the SPC users with quality tools, the open-source statistical software R [20] has important packages for building control charts and linking the academic and industrial communities. The main ones are: `qcc` [21], `qcr` [22], `qualityTools` [23], `SixSigma` [24,25], and `spc` [26]. It is possible to build several parametric and nonparametric control charts through these packages. In this manner, this work intends to present a new R package, called `npSPC`, containing the unprecedented implementation of some nonparametric control charts described in the book "Nonparametric Statistical Process Control" by Chakraborti & Graham [18]. In this way, it becomes simpler and faster to monitor processes whose data distribution is not known.

This study was also motivated by the need for monitoring tools to analyze the maintenance performance of two types of electrical boards from a mining company located in Atacama (Chile). These electrical panels play a crucial role in supplying and distributing electrical energy throughout the facilities. For instance, two major mining categories include surface mining and underground mining, in which the underground type greatly needs continuous energy to control temperature and oxygen flux, supported by these electrical boards.

The remainder of this paper is structured as follows. Section 2 presents the univariate (parametric and nonparametric) SPC charts in the R software, supported by some related works. Section 3 shows simulation cases and the advantages of nonparametric charts. Section 4 illustrates a real case adopted from an electrical mining company, describing its data set characteristics and comparing two electrical board maintenance plans. Section 5 contains the conclusion of this work. The appendix contains the mathematical details of the nonparametric hypothesis tests and their corresponding control charts.

2. Visualizing and monitoring processes

Often, the control charts presented in the literature, considering an invariant or rolling-window approach, are the parametric Shewhart (e.g., $\overline{X}$ and Range/S charts), CuSum, and EWMA types. The $\overline{X}$ chart is used to monitor the average of a process over time (in the same way the Range/S charts are used to monitor the variability of a process over time, by considering the range or standard deviation as the measure of dispersion). It is constructed by calculating the average of samples collected at different points in time and plotting these values on a two-dimensional graph. This kind of control chart allows for identifying trends or significant deviations in the process mean. Additionally, the Western Electric Company and Nelson [27] added some extra interpretation to the $\overline{X}$ chart with eight rules to detect shifts and patterns based on the invariant process characteristics. These rules try to capture local patterns, which later could be quantified through rolling process capability indices (PCIs) [17], which are not discussed in this work. The eight rules are:

(R1) Out-of-control state, if at least 1 point is greater than 3 standard deviations from the process mean;
(R2) Alert, if 2 sequential points out of 3 are greater than 2 standard deviations from the mean;
(R3) Alert, if 4 sequential points out of 5 are greater than 1 standard deviation from the mean;
(R4) Alert, if 8–9 points in-a-roll are placed on the same side of the process mean;
(R5) Alert, if there are 6 points in-a-roll increasing or decreasing;
(R6) Alert, if 14 points in-a-roll alternate increasing-decreasing pattern;
(R7) Alert, if there are 15 points in-a-roll within less than 1 standard deviation from the mean (either side);
(R8) Alert, if there are 8 points in-a-roll on both sides of the mean, but none are falling inside the range of 1 standard deviation from the mean.

On the other hand, the CuSum chart (also called the parametric CuSum score chart) is used to monitor the cumulative sum of deviations from a reference value. The mean-adjusted CuSum returns the scores equivalent to linear regression limits, but now as constant limits (horizontal lines) [17]. It is an effective alternative to Shewhart charts, which present "no memory", for detecting small changes in the process by accumulating differences between actual and expected observations. When the cumulative sum exceeds a predefined limit, it indicates a possible process change.

Finally, the EWMA chart is a variation of the CuSum chart that assigns greater weight to more recent data points (like a rolling window) [17]. It is beneficial when more importance is given to recent observations rather than treating all points equally. To present different SPC applicable scenarios, Table 1 summarizes previous works and different applications across the standard univariate SPC techniques. Other nonparametric approaches (not covered here), including adaptive methods and miscellaneous nonparametric control charting techniques, are discussed in Chapter 5 of Chakraborti and Graham [18]'s book.

According to Chakraborti and Graham [18], as in parametric control charts (or nonparametric approach), there are two possible cases in which applications can fit: case K, where the parameters are known or specified (using the data distribution information); and case U, where the parameters are unknown or unspecified (using another sample/information as a proxy). The difference between the two cases lies in evaluating/deciding whether there is a need to estimate the parameter of interest or not, and only adopting the observed data/sample (and usually split into phases 1 and 2, which will be better described/distinguished later).

2.1. The npSPC package

The R package developed and presented here is called `npSPC`. It is available on GitHub (https://github.com/CER-UFBA/npSPC) and contains 12 different functions for signs or signed-rank nonparametric control charts: six of them are related to the case of known in-control parameters, whereas the other six are related to the case of unknown parameters (Fig. 1). These functions are listed in Table 2 and will be further detailed later. Finally, the sign functions will be adopted whenever the comparison of two groups is to be tested, and the signed-rank functions when, in addition to the symmetry checking, the pair of observations is sampled with the order.

This new package can be installed in the R software [45] using the *install_github* function from the `remotes` package [46], as shown in the command below:

```
> remotes::install_github('CER-UFBA/npSPC')
> library(npSPC)
```

Moreover, the package information is available at https://github.com/CER-UFBA/npSPC. Further details about each function can be found in the README (and Table 2). Appendix contains the mathematical details of the nonparametric hypothesis tests and their control charts. Our proposed R package `npSPC` establishes the control limits

Table 1
Related works with univariate SPC control chart tools (*techniques in gray are parametric control chart tools).

Tecnique	Field	Motivation/Description
|Case K – Known parameter|		
$\overline{X}$ chart*	Materials science	Chronostatistics as a variogram lying in its simplicity and effective tool for identifying the components of process variability [28]
Range chart*	Medicine	Comprehensive quality Assurance (QA) programs for the whole variation of the radiotherapy process among patients [29]
CuSum*	Biomedical engineering	QA limits on the particular beam parameters and detecting offsets in the machinery readings [30]
EWMA*	Industrial	An automatic control system for improving the uniformity of a quality measurement on textured continuous-filament carpet yarn [31]
Shewhart Signs	Supplements manufacturing	Monitoring of CO_2 dissolved within the soft drink for polyethylene terephthalate (PET) bottles [32]
Shewhart Signed-Ranks	Educational field	To monitor the cumulative grade point averages (GPAs) of students over time and detect any statistically significant positive or negative shift from a desired target value [33]
CuSum Signs	Industrial	To detect a small shift or change in the process median using a well-known set of data from [34, Tables 5.1 and 5.2] on inside diameters of piston rings for an automotive engine produced by a forging process using the data quantile information [35]
CuSum Signed-Ranks	Materials science	Comparison of observed pairs of (identical) replicate coal ash, analyzed by two independent laboratories, adopting a distribution-free and scale-invariant scheme [36]
EWMA Signs	Civil Engineering	Water consumption monitoring, detecting water leaks in residential apartment buildings and identifying changing patterns [37]
EWMA Signed-Ranks	Medical-IoT	Monitoring online vibration data for predictive maintenance applications [38]
|Case U – Unknown parameter|		
Shewhart with precedence statistic	Industrial engineering	A monitoring scheme of a production milk bottles, where the quality characteristic of interest is the volume (in ml) of milk in each bottle [39]
Shewhart with Mann–Whitney statistic	Mechanical engineering	Piston rings quality effective performance in spotting deviations in the process location for heavy-tailed and skewed distributions [40]
CuSum with exceedance statistic	Industrial	To detect smaller shifts in the median of the distribution of inside diameters of piston rings manufactured by a forging process using another sample as Ref. Mukherjee et al. [41]
CuSum with Wilcoxon rank-sum	Industrial production	To maintain the amount of soft drink contained in the bottle within certain limits from the real data set of Pepsi Cola production company in Al-Khobar, Saudi Arabia [42]
EWMA with exceedance statistic	Industrial	To detect smaller shifts and when the median pattern, under time-dependence resolution, of the changes in the diameters of piston rings manufacturing process [43]
EWMA with Wilcoxon rank-sum	Mining industry	Monitoring the percentage of silica concentrate that remains as an impurity at the end of the froth flotation process [44]

Table 2
Implemented functions in the proposed `npSPC` R package.

Command	Control chart function description	Hyperparameters
|Case K|		
shewhart_sn	Shewhart based on signs	med = 0, far = 0.0027
shewhart_sr	Shewhart based on signed-ranks	mu0 = NULL, far = 0.0027
cusum_sn	CuSum based on signs	med = 0, k = 0.5, h = 5
cusum_sr	CuSum based on signed-ranks	med = 0, k = 0.5, h = 5
ewma_sn	EWMA based on signs	med = 0, lambda = 0.05, L = 2.472
ewma_sr	EWMA based on signed-ranks	med = 0, lambda = 0.05, L = 2.472
|Case U|		
shewhart_prec	Shewhart based on precedence statistic	j = n/2, far = 0.0027
shewhart_mw	Shewhart based on Mann–Whitney statistic	cl = (n1*n2/2 + 1), far = 0.0027
cusum_ex	CuSum based on exceedance statistic	r = n/2, k = 0.5, h = 5
cusum_wr	CuSum based on Wilcoxon rank-sum statistic	r = n/2, k = 0.5, h = 5
ewma_ex	EWMA based on exceedance statistic	lambda = 0.05, L = 2.447, r = n/2, far = 0.0027
ewma_wr	EWMA based on Wilcoxon rank-sum statistic	lambda = 0.05, L = 2.447

defined by the hyperparameters, considering by default the values presented in the book "Nonparametric Statistical Process Control" [18], where, among others, `n` is the sample size, `med` is the median, and `far` is the false alarm rate.

After choosing the appropriate control chart to meet the monitoring goals, the control limits should be carefully determined. In SPC approaches, a stable phase of the process, known as Phase I, is used to estimate/learn these control limits (upper and lower bounds). Then,

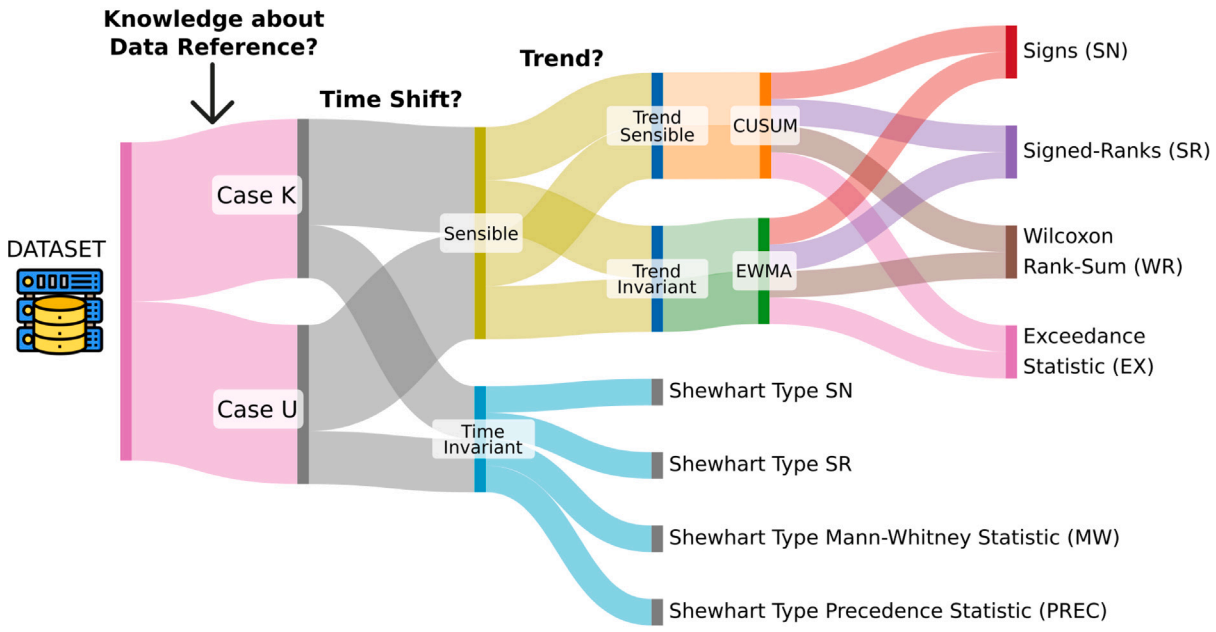


Fig. 1. Sankey diagram describing the working flow of the npSPC package and its implemented class of functions.

Phase II is applied, adopting new data to affirm the quality of those estimations. At the end of Phase I, the in-control (IC) model should be established.

3. Simulated scenarios

In this subsection, we present three mining-based applications, adopting the focus of comparing classic parametric tools versus some nonparametric approach, mainly: (i) the CuSum chart based on signed-ranks versus the parametric CuSum; (ii) the EWMA chart based on exceedance statistic versus the parametric EWMA; and (iii) the Shewhart chart based on signed-ranks versus the $\overline{X}$, its parametric version. These examples showcase some possible applications of the new package and compare the behavior of the control charts available in it with parametric versions and their opposing points.

3.1. Nonparametric CuSum charts

The first data set is related to identifying and quantifying the variography of the process rule between the proportion of a Target Material and the Observation Sample, which shall be treated as random variability and must be minimized. The fictitious data set contains $m = 20$ samples (collections as a block); in each block, observations (replicas) sized $n = 4$ were taken, and the difference between the Target value and Sampled is a continuous value and expected to follow a symmetric distribution, with a specified median of 0 (illustrated in Fig. 2).

If the sampling process is considered independent across the blocks, then an SPC tool "without memory" is enough to explain the process dynamics. However, if the blocks are somehow related, Shewhart's $\overline{X}$ or S charts will not be helpful.

In the mining process, e.g., incremental sampling of flowing streams is shown to be samples collected at regular intervals from a moving conveyor belt [28]; samples are said to be "not independent from one another". Moreover, there was a strong correlation between one sample and the next, and the strength of the correlation declined as the samples became further apart. In this manner, the presented proportion of a Target Material and the Observation Sample can be seen as a quality assurance of a batch, and adopting the parametric CuSum and EWMA charts looks useful in quantifying the dynamics of this proportion.

Table 3
Simplifying the nonparametric CuSum statistic reasoning by taking each block median plus the interquartile difference between the current and previous blocks. The block position ordered estimation is presented, and, in bold, the ones statistically significant.

Position (#)	**12**	**10**	13	9	6	⋯	19	7	1
Value	**2.018**	**1.813**	1.615	1.490	1.350	⋯	0.043	0.010	0

On the other hand, the nonparametric CuSum chart, based on signs (CuSum-SN) or signed-ranks (CuSum-SR), is competitive to compare each ith block median added with the difference between the block dispersion $(i-1)$th and ith. The results show that the nonparametric CuSum-SR control chart signaled two points (#10 and #12) outside the established limits, indicating potential material stream median variation (Fig. 3). In contrast, all parametric charts ($\overline{X}$, S, CuSum, and EWMA) did not recognize the out-of-control patterns.

The interpretation of the CuSum-SR is that, as the sample collection goes, the block median representation plus the dispersion gap between the current state and the previous one is considered. That is, between block samples #10 and #9, a statistically significant dynamic was noticed, as well as a significant difference in the block samples #12 and #10 (Table 3).

That is, the biggest (statistically meaningful) median-dispersion relationship observed was between samples #12 and #11, and the smallest (statistically meaningful) median-dispersion trade-off observed was between samples #10 and #9. Moreover, these pairwise one-step forward comparisons can be seen in Fig. 2.

3.2. Nonparametric EWMA charts

Another important gap in the mining company is to reach equivalent conditions in pyrometallurgical processes for primary production, e.g., spectral analysis involved in copper pyrometallurgy, performing control strategies of the smelting and converting processes [47]. Considering the continuous part of the spectrum, the calculation for the flame temperature is possible by applying the Planck radiation law, though an adiabatic temperature model has been proposed based on the mass and energy balance; later, this model can be obtained. The observed random variable is the difference between the measured

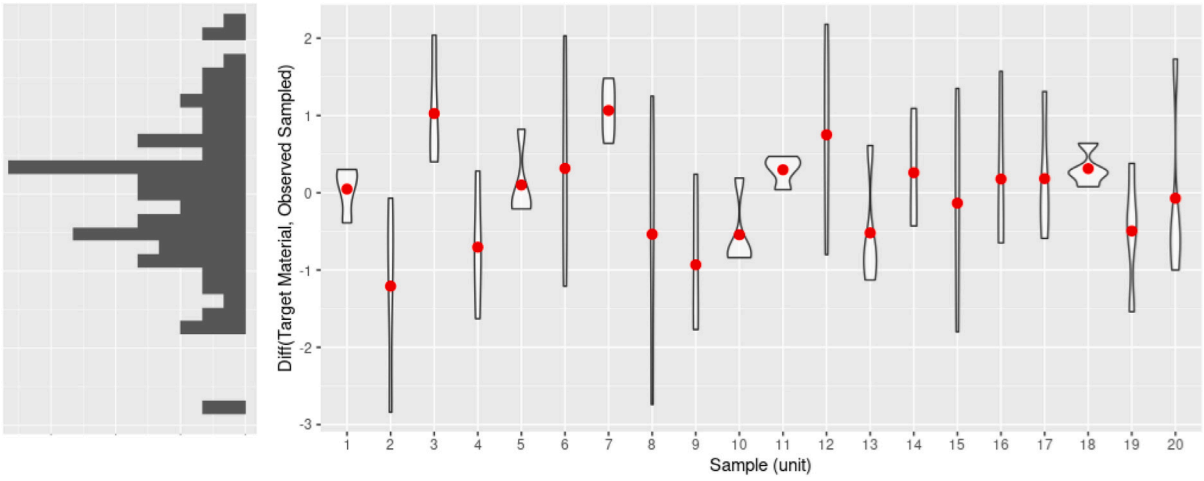


Fig. 2. Difference between target material and observed sampled material. The red dots are the median of each block/sample. The violin plot represents the dispersion of each block.

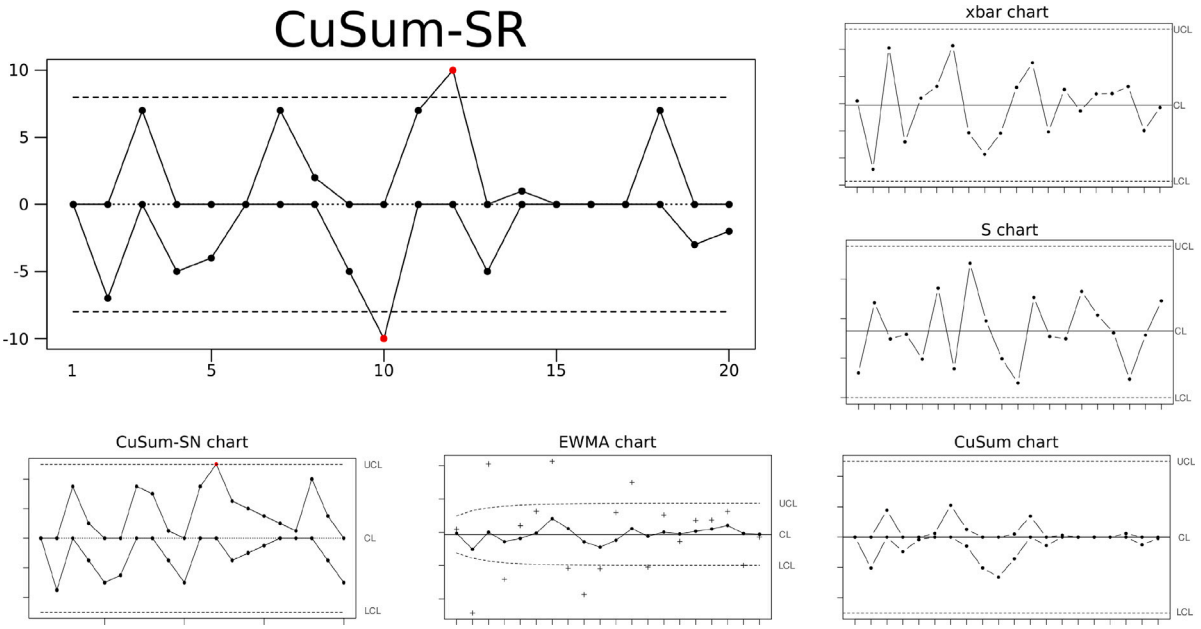


Fig. 3. Control charts (with the sample number on the x-axis and the corresponding sample statistic on the y-axis) for the first data set: CuSum-SR chart (top-left); $\overline{X}$ chart (top-right); S chart (center-right); CuSum chart (bottom-right); EWMA chart (center-bottom); and CuSum-SN chart (left-bottom).

(radiometric experiments) and calculated temperature for laboratory combustion of concentrates using a mass and energy balance. The fictitious data set contains $m = 15$ samples of size $n = 10$, coming from a process with an unknown, continuous, and symmetric distribution, with a median equal to 0 (Fig. 4).

An important fact to notice is that a single positive value was obtained in the second block. This dragged the block mean down (not much its median, compared to the next and previous blocks). Nonetheless, other blocks show greater dispersion and values below the minimum from block #2 (Fig. 5).

Fig. 6 reveals the estimated dynamic pattern based on SPC tools (parametric $\overline{X}$, S, CuSum, and EWMA charts, as well as nonparametric EWMA charts). The process shows to be stable (in-control), although the $\overline{X}$ chart showed to be sensitive towards the pattern of block #2, which suggests an out-of-control condition.

This example indicates how cautious the user should be when choosing the mean versus the median process representation. Another important SPC tool is the average run length (ARL), which indicates the expected number of out-of-control samples in several observations [17]. Considering the EWMA-SN algorithm (with parameters $\lambda = 0.05$ and $L = 2.612$) resulting in $\text{ARL}_0 \approx 500$, within each replica size of $n = 10$. Whereas, the ARL_0, under the supposition of normality, based on the $\overline{X}$ chart, would expect approximately 1 out of 370 samples to be out-of-control (false positive), that is,

$$\text{ARL} = \frac{1}{p}, \quad \text{where:}$$

$$\mathbb{P}(\mu - 3\sigma_{\overline{x}} < p < \mu + 3\sigma_{\overline{x}}) = \boldsymbol{\Phi}\left(\frac{\text{UCL} - \mu}{\sigma_{\overline{x}}}\right) + \left[1 - \boldsymbol{\Phi}\left(\frac{\text{LCL} - \mu}{\sigma_{\overline{x}}}\right)\right] \approx 0.0027,$$

then

$$\begin{cases} \hat{\mu} = \overline{x} = 0.074 \\ \hat{\sigma}_{\overline{x}} = \frac{\hat{\sigma}}{\sqrt{n}} = \frac{1.009}{\sqrt{10}} \approx 0.319 \end{cases} \rightarrow \begin{cases} \text{UCL} = 1.031647 \\ \text{LCL} = -0.88298 \end{cases} \rightarrow \widehat{\text{ARL}_0} = \frac{1}{\hat{p} = 0.00269} \approx 370,$$

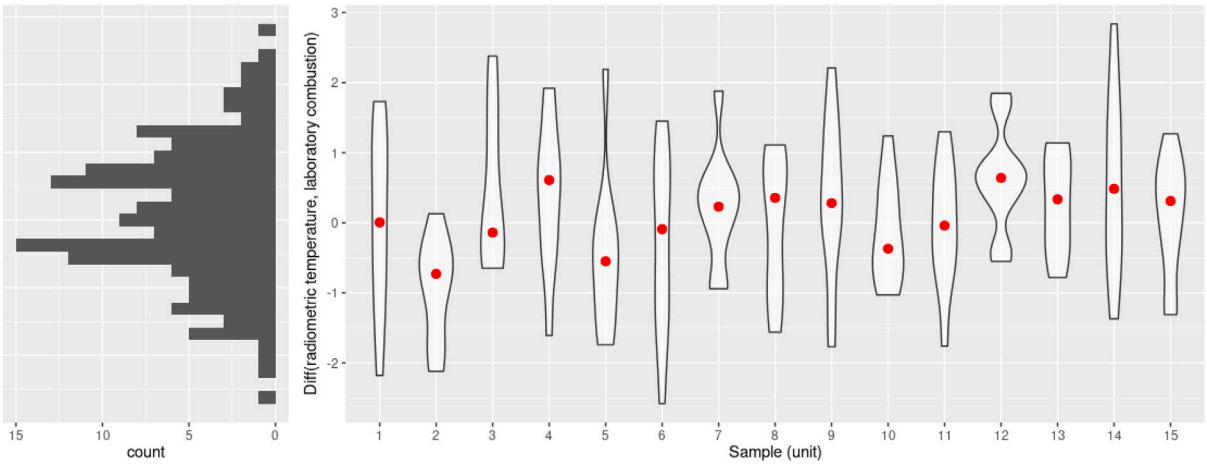


Fig. 4. Difference between Radiometric experiments and Laboratory calculated temperature. The red dots represent the median of each observed block, and the violin is the block dispersion.

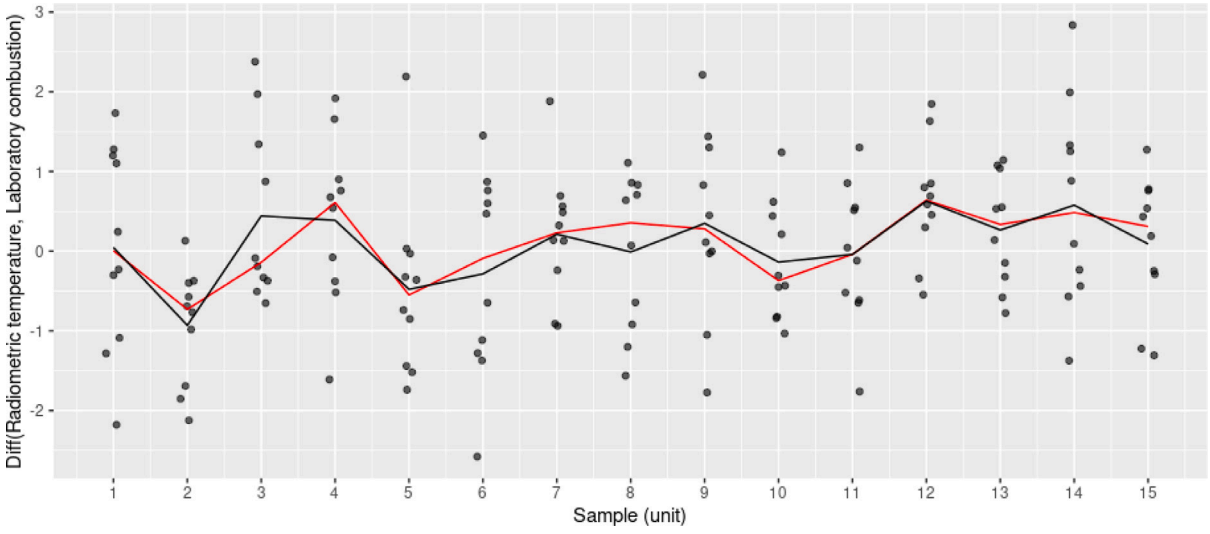


Fig. 5. The dot plot of each observation (per block). The black line is the process mean, and the red line is the median.

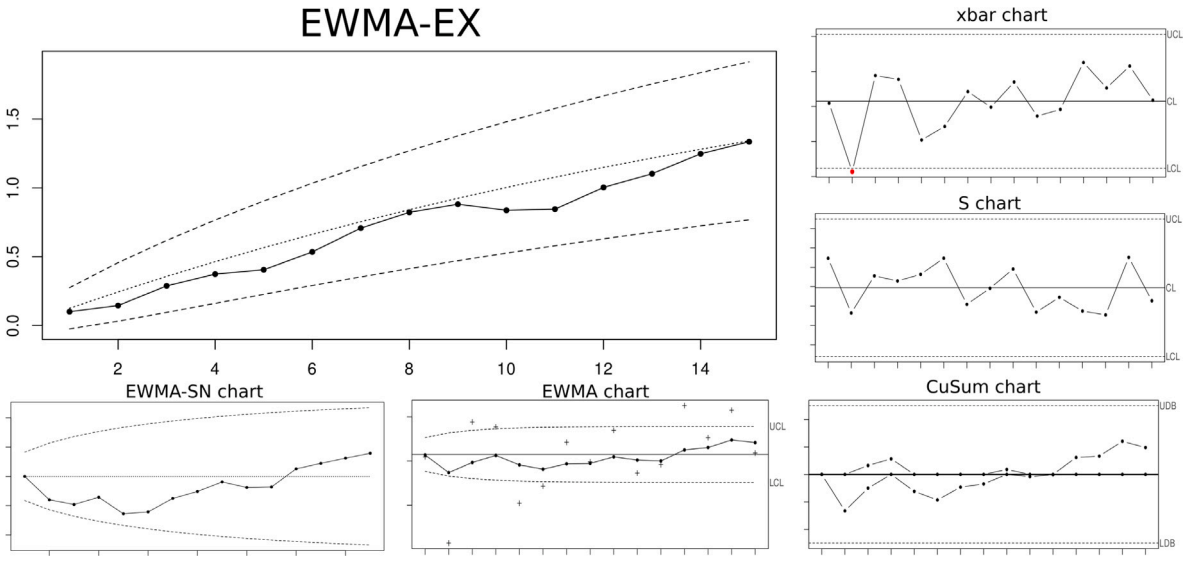


Fig. 6. Control charts (with the sample number on the x-axis and the corresponding sample statistic on the y-axis) for the second data set: EWMA-EX chart (top-left); $\overline{X}$ chart (top-right); S chart (center-right); CuSum chart (bottom-right); EWMA chart (center-bottom); and EWMA-SN chart (left-bottom).

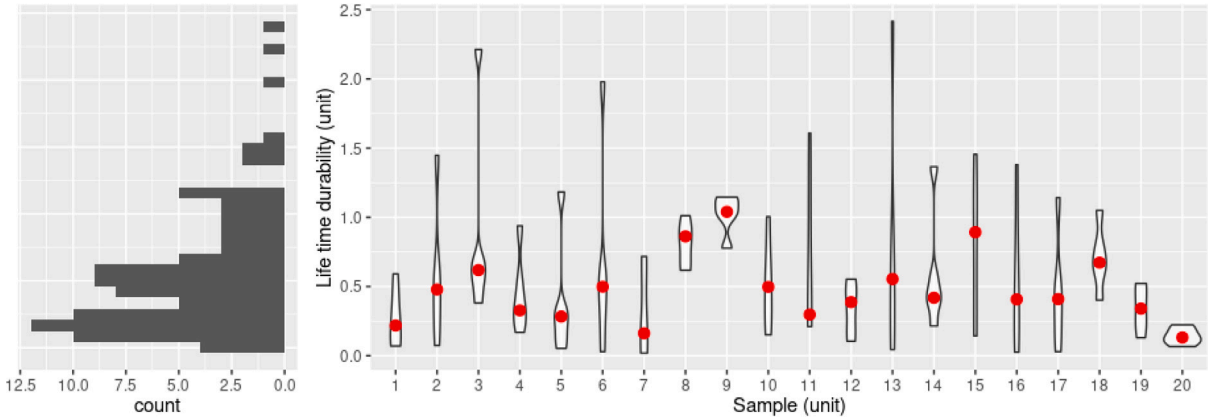


Fig. 7. Product lifetime observation following an exponential distribution. The red dots represent each block median, and the violin their dispersion.

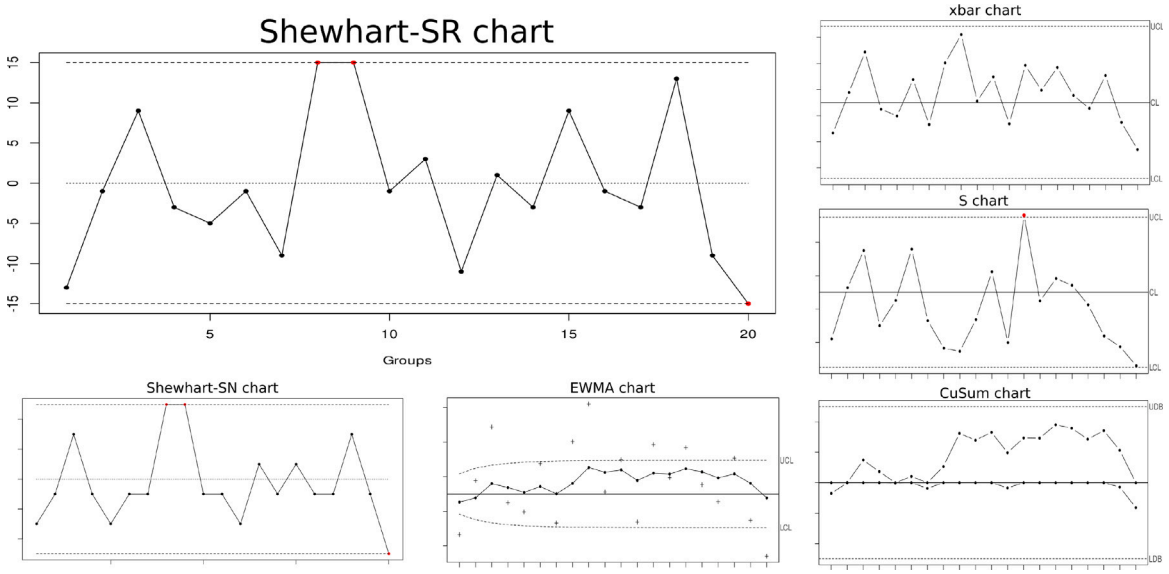


Fig. 8. Control charts (with the sample number on the x-axis and the corresponding sample statistic on the y-axis) for the third data set: Shewhart-SR chart (top-left); $\overline{X}$ chart (top-right); S chart (center-right); CuSum chart (bottom-right); EWMA chart (center-bottom); and Shewhart-SN chart (left-bottom).

where $\Phi(\cdot)$ is the standard normal cumulative density function. Therefore, the assumption of normality may bring false risk (information) to estimate the expected number of out-of-control observations based on the Six Sigma tolerance, given that the process is in control (ARL_0).

3.3. Nonparametric Shewhart charts

The third example was constructed using a recurring motivation in the engineering field, adopting the lifetime observation task. It is widely discussed that an object's lifetime may follow an exponential distribution. Therefore, a synthetic set with a mean equal to 0.5 was created. That is, the center of the process is shifted to 0.5 units. Fig. 7 illustrates a situation of non-normality (where nonparametric control charts may be more suitable [48]), which considers $m = 20$ blocks with replica size $n = 5$. Additionally, the samples #8 and #9 were forced to be out-of-control.

Fig. 8 presents the parametric $\overline{X}$, S, CuSum, and EWMA charts, as well as the nonparametric Shewhart's (SN and SR) charts. The S chart captured block #13 as a statistically significant dispersion range according to the standard deviation. Nonetheless, blocks #8 and #9 were smoothly unnoticed, as well as the last block #20 (with low dispersion and median). On the other hand, the block pattern switch (#8 and #9) was only noticeable through the nonparametric charts.

Table 4
Computational cost - `qcc` versus `npSPC` package.

Sample (n)	Time (s)					
	$\overline{X}$	Shewhart-SN	CuSum	CuSum-SN	EWMA	EWMA-SN
30	0.0206	0.0151	0.0329	0.0164	0.0330	0.0236
500	0.0546	0.0462	0.0309	0.0272	0.0409	0.0374
5000	0.1012	0.0524	0.1084	0.0227	0.1024	0.0337
50,000	0.5080	0.2644	0.6073	0.0242	0.6152	0.1032
5,000,000	48.7703	32.4167	58.6741	8.8503	71.8580	5.1880

All the parametric control charts ($\overline{X}$, S, CuSum, and EWMA) failed to identify the out-of-control condition of this process. Whereas the nonparametric Shewhart's (SN and SR) charts captured both out-of-control samples (#8 and #9) and the abnormal median-dispersion sample #20. In both cases, nonparametric Shewhart's (SN and SR), the specified ARL_0 was close to 370.

Table 4 shows the computational cost, in seconds, considering artificial data sets with 30, 500, 5000, 50,000, and 5,000,000 observations. Overall, the proposed nonparametric algorithms showed a 51.60% average operational execution time less than the `qcc` parametric algorithms.

In the next section, we will present a real-case mining company electrical monitoring task in which two types of electrical boards will be



Fig. 9. Illustration of the electrical board types inside the mining company in Atacama (Chile).

compared. The importance of monitoring these equipments are related to the oxygen and electrical power distribution inside the mining, which is essential for operation.

4. Results - mining company case

Implementing an appropriate preventive maintenance plan, in mining field, includes three types of maintenance which can help minimize the chances of failures in electrical equipments and comply with the requirements of the Chilean Technical Regulatory Specification (RIC-N°17.6). The types of maintenance can be: (i) time-based preventive maintenance (scheduled at regular intervals); (ii) condition-based preventive maintenance (scheduled based on the actual condition of the equipment); and (iii) predictive maintenance (statistical analysis to predict failures before they occur). For the predictive maintenance plan, historical data must be analyzed, which requires the use of an appropriate SPC monitoring tool; and the goal of this case study is to demonstrate how to suggest a preventive maintenance plan to reduce the proportion (risk/probability) of items being removed in a company located in northern Chile. From a supply chain perspective, the company's reliability-centered maintenance and inventory should be planned to ensure that the mining operation can provide electrical power distribution for all activities, with the necessary items (types) available when needed.

4.1. Electrical distribution in a mining company

The adopted data set includes registers from 157 electrical panels, divided into two categories (TDF and TDF/SM), which quantify the amount of maintenance, whereas the item could be operational after it or removed for greater repair. The failure recurrence is registered and subsequently determine its risk/preventive odd. Fig. 9 illustrates some of the mentioned electrical boards products, which are essential to keep the miner operational and habitable, thus requiring its quality assurance. The observed data span from January 2019 to May 2023, resulting in 8321 observations (monthly records).

The most important variable observed was the monthly status of the board, where each month it could suffer maintenance, then: (i) stay in operation; or (ii) be removed (with or without the possibility of returning to operation). The response variable (or target feature) is binary, sometimes called a dummy variable, which is the random variable, Y, representing the two maintenance outcome groups (0 = "in operation" or 1 = "removed item"). The maintenance temporal distribution (per type) across the years is presented in Table 5.

Table 5
Summary of the total maintenance records related to the two electrical board types (per year).

	Year	Total maintenance				# Removed
		0	1	2	> 2	
TDF	2019	615	151	18	140	144 (15.6%)
	2020	766	35	32	91	144 (15.6%)
	2021	795	30	42	57	144 (15.6%)
	2022	789	36	72	27	144 (15.6%)
	2023	278	35	71	1	60 (15.6%)
TDF/SM	2019	452	171	0	337	36 (3.75%)
	2020	574	97	6	283	36 (3.75%)
	2021	683	26	12	239	36 (3.75%)
	2022	714	67	20	159	36 (3.75%)
	2023	317	10	16	57	15 (3.75%)

Based on historical follow-up records of 157 electrical panels, 77 items are related to the TDF category, and 80 items are related to the TDF/SM one. The total amount of maintenance executed/observed was 8321 records (monthly interventions), whereas 4081 were TDF-type (3445 remained operational and 636, or 15.58%, items were removed and replaced due to operational issues) and 4240 were TDF/SM-type (4081 remained operational and 159, or 3.75%, items were removed). This dynamic can be observed in Fig. 10, where a noticeable decrease in the total monthly predictive maintenance actions for both electrical board types is shown (solid lines), along with an increase in the number of removed items (replacements, shown with dashed lines).

As each electrical board type contains a sequential follow-up, the first analysis was towards the monthly median of the removed items. Therefore, the boards (per month) can be seen as replicas of the same process, given their equivalent characteristic. Fig. 11 shows the comparison of the median items per electrical board type across time.

Since the high-risk is related to the number of removed items (TDF versus TDF/SM), the Student's t-test was used to compare the means between these two groups, and the U-test was used to compare their medians. Table 6 shows the statistics of the tests, in which both reject the equality of the groups' means or medians. Both tests suggest that the electrical board type shows different dynamics towards reliability-centered maintenance.

SPC control chart tools help analyze where the process shows out-of-control patterns across time. Although the random variable, Number

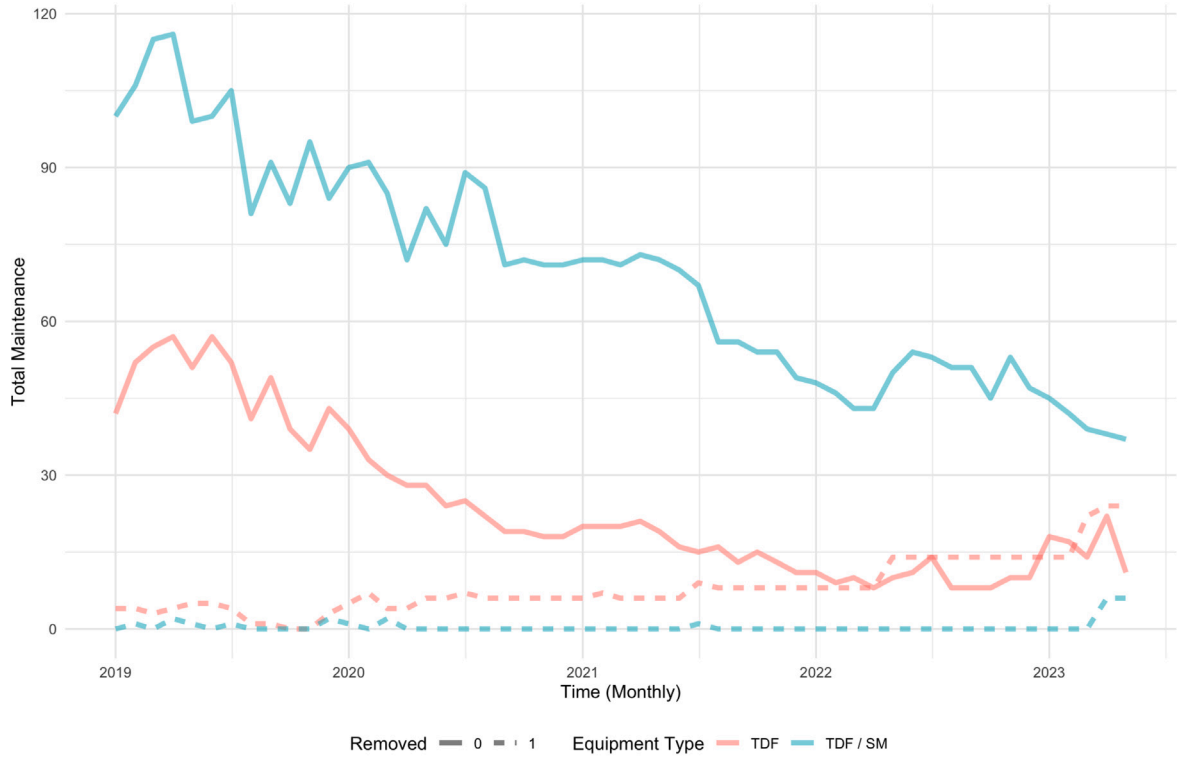


Fig. 10. Electrical boards' monthly maintenance number and their operationalization (also removed items per type). The solid lines are the executed maintenance leaving the board operational, while the dashed lines indicate the dynamic of maintenance with items removed.

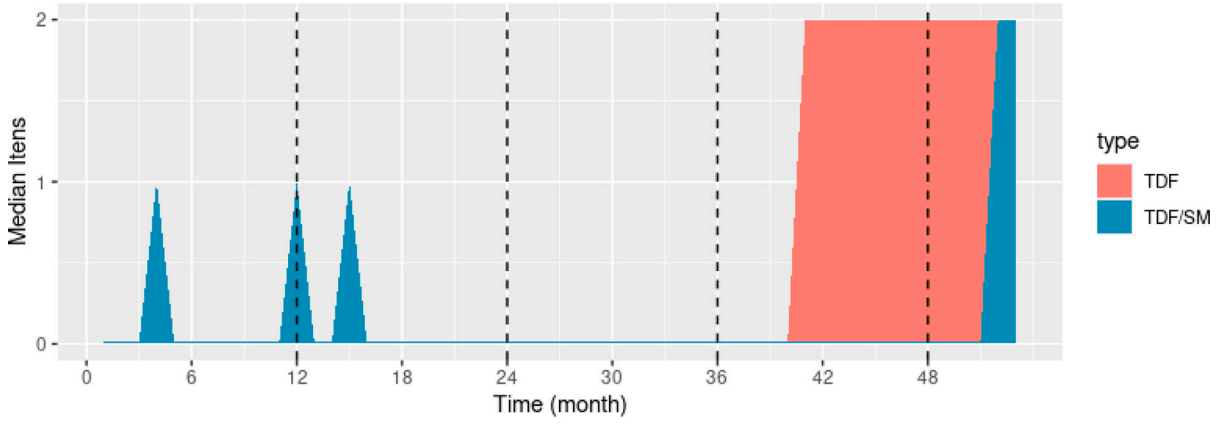


Fig. 11. Item-removed median evolution per electrical board type (TDF and TDF/SM).

Table 6
Student's t-test and U-test between the central measures of TDF and TDF/SM, along with normality tests for each variable (electrical board type). $CI_{95\%}$ = 95% confidence interval for the mean difference.

	Statistic	*p*-value	$CI_{95\%}$
Student's t-test	−10.463	2.2×10^{-16}	[−0.63, −0.43]
Mann–Whitney U test	36,596	2.86×10^{-11}	–
Shapiro–Wilk normality test (TDF/SM)	W = 0.356	2.2×10^{-16}	–
Kolmogorov–Smirnov test (TDF/SM)	D = 0.5	2.2×10^{-16}	–
Anderson–Darling normality test (TDF/SM)	A = 48.164	2.2×10^{-16}	–
Shapiro–Wilk normality test (TDF)	W = 0.629	2.2×10^{-16}	–
Kolmogorov–Smirnov test (TDF)	D = 0.5	2.2×10^{-16}	–
Anderson–Darling normality test (TDF)	A = 115.62	2.2×10^{-16}	–

of Maintenance, is countable (not symmetric), parametric tools were adopted for the sake of illustration. Fig. 12 presents the estimated behavior of four parametric control charts applied to maintenance with removed items. The first highlight is that the process-average was considered centered at 0 (as expected to show zero fail cases). The left-hand charts are related to TDF/SM, and the right-hand charts to

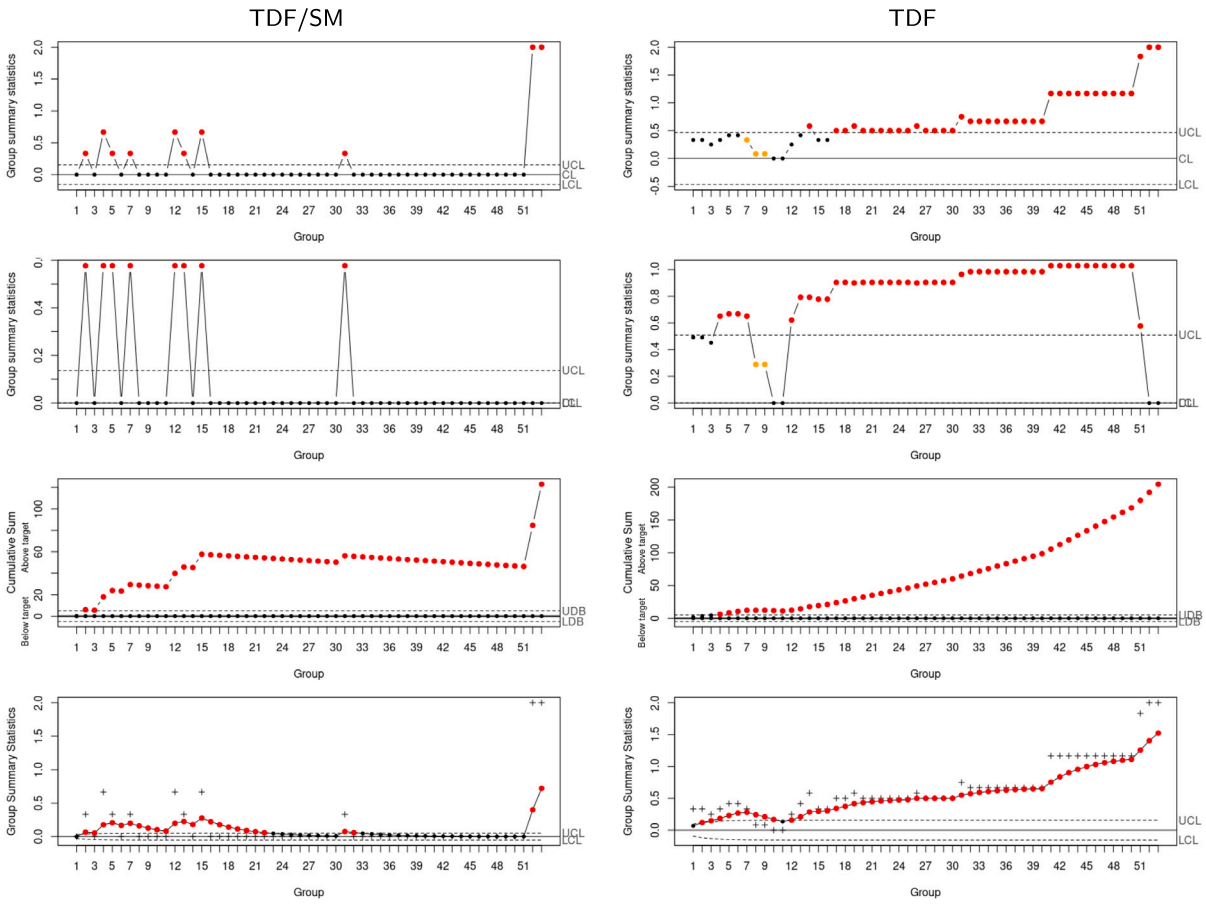


Fig. 12. All parametric SPC charts: $\overline{X}$ charts (first-line); S charts (second-line); CuSum charts (third-line); and EWMA charts (fourth-line). The left column is related to TDF, and the right is related to the TDF/SM board type.

TDF board type. Regardless of the tool ($\overline{X}$ charts (first-line); S charts (second-line); CuSum charts (third-line); and EWMA charts (fourth-line)), many out-of-control points are visualized, suggesting a "messy" process and making it difficult to identify patterns using a normal-based reference distribution.

The electrical board types are both presenting out-of-control samples, though the TDF suggests stronger out-patterns in the first year of functionality (presented by the $\overline{X}$ and S charts under the normal-distribution supposition). Nevertheless, the CuSum chart of both processes shows high time dependence, suggesting a periodic maintenance plan. However, an important characteristic to be aware of is that the process mean can be easily influenced and contaminated due to the presence of outliers (both trends confirmed through Fig. 10 dashed lines).

Another point against the parametric (normal-based) SPC tools, especially $\overline{X}$ and S charts, is their unclear interpretation of the control limits. Moreover, the $\overline{X}$ chart centered at 0 shows an upper control limit (UCL) of ≈ 0.15 for the TDF set (and ≈ 0.47 for the TDF/SM), which is the tolerable removed item per block (these values, in practice, bring harder interpretation for the decision-making). However, this limit will be corrected by adopting a nonparametric approach.

Fig. 13 presents the nonparametric SPC charts: (i) Shewhart-SN; (ii) CuSum-SN; and (iii) EWMA-SN, considering the median as a reference for the maintenance with removed items. Since the presence of out-patterns (skewed information) is commonly recurrent, each monthly mean is easily contaminated; therefore, the normal-based SPC statistic is associated. For instance, the Shewhart-SN chart suggests only an out-of-control median pattern in the last two months for both electrical board types. This information is also supported by the last observed semester (#49-53), in which the lowest level of maintenance was observed (regardless of removing items or not).

Despite the Shewhart-SN results/characteristic, the nonparametric CuSum-SN and EWMA-SN charts also suggest that a temporal influence is carried. The CuSum-SN chart presents the overall picture, pointing out the statistically significant abnormality amount of maintenance greater than the desired zero median (confirmed in Table 5). The EWMA-SN chart shows that this out-pattern is not common every month (black dots), which is looking at closer temporal windows.

Regardless of the mean or median approach, much evidence shows the instability of the observed process (therefore, making it difficult to define a stable period for calculating the control bounds). Moreover, it is unfair to use the asymmetric data to estimate the SPC bounds-of-limits, which, in this case, is only desired to know if the Removed-for-Maintenance group presents a relational pattern to the In-operation. Therefore, it is even more challenging to split the data set into the partition phases 1 and 2 (aiming for the parameter estimations of the bound limits). Then, one falls into the unknown-parameter case, where it is preferable to adopt a bivariate analysis rather than a univariate approach alone, and then compare the dynamics between non-removed items (as a reference/proxy) and the removed ones.

A deeper look at ranked signs can unveil this temporal dependence, adopting the EWMA since the historical data exhibit a time-shift sensitivity. Fig. 14 shows the nonparametric EWMA-WR charts, which use the predictive-maintenance to explain the corrective-maintenance with removed items (that is, non-removed electrical board type as reference, like phase 1 to estimate parameters of the removed items pattern). The literature calls "nonparametric SPC - case U". Moreover, the results showed the period where the median pattern between the Removed

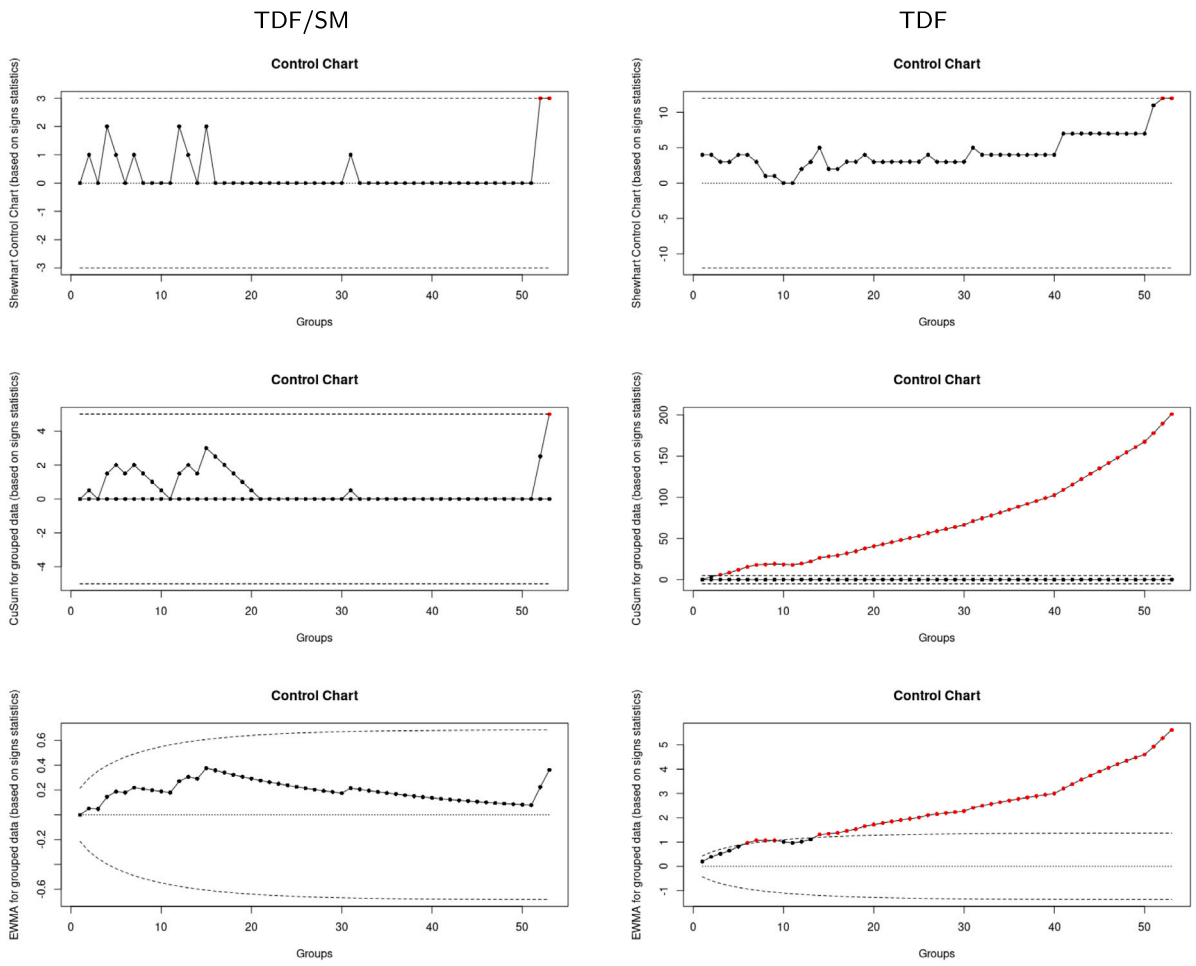


Fig. 13. All nonparametric SPC charts (case K): Shewhart-SN charts (first-line); CuSum-SN charts (second-line); and EWMA-SN charts (third-line). The left column is related to TDF, and the right is related to the TDF/SM board type.

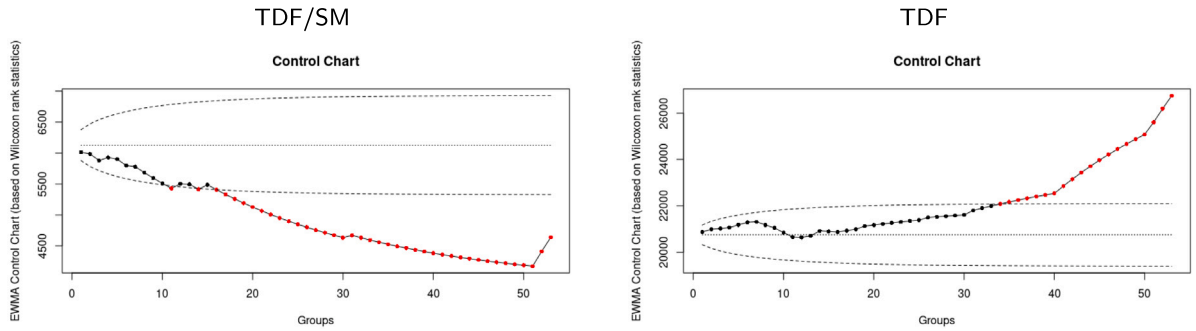


Fig. 14. The EWMA-WR charts for the TDF (left-hand) and TDF/SM (right-hand). In both cases, it was used the non-removed items were used as a parameter estimation proxy (case U).

and Operation groups differs within the starting data marked with the red dots.

The TDF electrical board type behaves equally in the first year (removed and non-removed items as a proxy, suggesting that the number of maintenance was meaningful at some point). Then, it seems the company has a different decision towards the maintenance/operational actions across the observed time. On one hand, the TDF electrical boards were affected approximately after the third year (after month #34 - Oct '21), whereas the TDF/SM after the month #16 (Apr '20). Fig. 15 shows the suggested minimum maintenance number, combining the preventive historical information and the SPC via EWMA-WR charts.

Wrapping up all the elements, monitoring the electrical boards' maintenance process can be improved in this company, especially with preventive maintenance strategies. The relationship between preventive and corrective maintenance operations exists (that is, removed and non-removed items), suggesting a deeper look at the risks of non-operation/removed electrical boards and financial/logistical costs

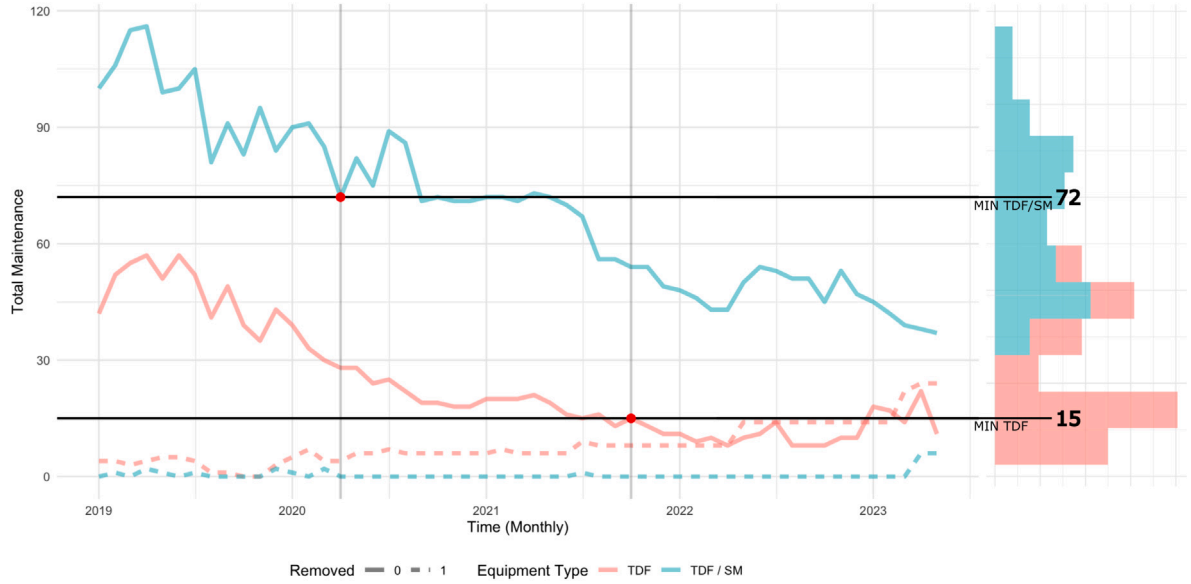


Fig. 15. Electrical boards' minimum monthly maintenance number given the temporal shift presented at the EWMA-WR charts for the TDF/SM (#16 - until April '20) and TDF (#34 - until October '21).

for their replacement. Furthermore, it suggests the minimum monthly maintenance number, if one considers a temporal influence (or even an imperfect repair scenario) through the preventive maintenance (board within operation) outcome as a proxy. The suggested scenario that could be drawn for reducing the number of removed items is a monthly total preventive maintenance of 72 TDF/SM-type electrical boards, and 15 TDF-type boards.

5. Conclusion

An industrial quality control strategy is a complex process that needs to be appropriately adapted for each scenario, which can lead any enterprise to temporal advantages. This work used some mining company tasks to illustrate the complexity of the decision-making based on the mean-median dynamic observation. In this manner, we discussed (and implemented) elements that can light up an industrial parametric versus nonparametric monitoring process.

As a first result, we developed a user-friendly R package for building univariate nonparametric control charts. This expands the range of nonparametric SPC techniques that can be implemented easily and quickly through this essential statistical open-source software with computational efficiency. This implementation links the importance of the SPC control chart methodology for data visualization process monitoring and potentially to the Industry 4.0 era [49].

Additionally, different real-world mining tasks were illustrated, showing cases of temporal dependence observation or sample contamination, which can affect the adoption of mean-based SPC tools. Moreover, the Big Data (sensors) era brings extra complexity, and normal data supposition can be dangerous to adopt. Alternatively, median-based (nonparametric) SPC tools are shown to be more robust and effective in the out-of-control patterns.

Big Data & Data Analytics are strongly related to Industry 4.0 [50], which is characterized by a transformation towards a society of hyperconnectivity, hyperintelligence, and convergence [51]. The concept emerges, presenting a significant challenge of automating the monitoring or control of industry quality and processes. Its advantages include increased competitiveness and performance, greater versatility and resilience, and higher profitability. An interconnection of devices that extends beyond the walls of the manufacturing plant, precise monitoring, tracking, receiving, and sending of raw materials [52], which can be assessed by the SPC algorithms [19].

A potential limitation of this study is related to the discussion on kernel functions and bandwidth selection methods; nevertheless, this work focused on the implementation of the existing methods mostly presented by Chakraborti and Graham [35]. The discussion towards the automatization of the hyperparameters for all these control charts is still vague (solutions are related to the ARL and Detecting Process Shifts; some literature can be found at Abbas et al. [53,54]); nevertheless, the proposed package also contains ARL functions implemented to develop such a discussion. Future studies shall explore simulation scenarios with different granularities, and data frequency shall be tested (which is now easier given the implemented tools). Another gap to be tested in future studies is the adoption of Conformal Inference with Machine Learning (ML) techniques to estimate the bandwidth control charts (the first elements can be found in [55,56]), then compared with these nonparametric approaches.

The elements of modernizing production processes incorporate intelligent devices such as machines, equipment, and instruments; intelligent systems applied in transportation, logistics, inspection, testing, education, Research and Development (R&D); intelligent automation to integrate all aspects of humans, machines, and software; and innovative materials [57]. The goal of technologies will evolve towards the concept of Industry 5.0, which must create an intelligent healthcare environment with real-time capabilities [58]. However, this study did not consider the multivariate scenarios often recurrent in real-world decision-making [59]. Nevertheless, this study is helpful in the discussion as a first look into the nonparametric SPC tools and their implementation. Future work shall extend in this direction, including feature analysis/selection, symbolic representation, and multivariate response. For studies on nonparametric multivariate control charts, the reader is referred to Zou and Tsung [60], Boone and Chakraborti [61], Li et al. [62], Mahmood and Erem [63] and Erem and Mahmood [64], among others.

CRediT authorship contribution statement

Diego C. Nascimento: Writing – review & editing, Visualization, Validation, Methodology. **Anderson O. Fonsêca:** Writing – original draft, Software, Methodology, Investigation, Formal analysis, Data curation. **Marcelo B. Fonsêca:** Writing – original draft, Software, Methodology, Investigation, Formal analysis, Data curation. **Moisés**

A.F. Silva: Writing – original draft, Software, Methodology, Investigation, Formal analysis, Data curation. **Paulo H. Ferreira:** Writing – original draft, Validation, Supervision, Methodology, Conceptualization.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used Grammarly to enhance their English grammar writing. After using these tools/services, the authors reviewed and edited the content as needed and took full responsibility for the content of the publication.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix

This appendix presents the mathematical details and definition of the control chart functions – Shewhart, CuSum, and EWMA – for both cases of known and unknown in-control limit specifications.

A.1. Case K (known in-control limits)

The control charts for case K, where the parameters are known or specified, implemented were: Shewhart based on signs, Shewhart based on signed-ranks, CuSum based on signs, CuSum based on signed-ranks, EWMA based on signs, and EWMA based on signed-ranks.

A.1.1. Shewhart control charts

Shewhart control charts [65] are the most popular methodology in SPC, mainly due to their simplicity. Comprised of three lines, namely the Lower Control Limit (LCL), Center Line (CL), and Upper Control Limit (UCL), they allow for checking whether the studied/analyzed process is under statistical control; for this purpose, the main requirement is that the plotted points fall within the control limits [17].

Shewhart based on signs

Consider the situation where samples of $n \geq 1$ observations are randomly taken at regular time intervals from a process with a continuous distribution and parameter of interest μ (here, the median). Suppose further that the known or specified value of μ under control is μ_0. Let $X_{i1}, X_{i2}, \ldots, X_{in}$, $i = 1, 2, \ldots$, be the sample at the ith time point, and suppose each observation X_{ij}, $j = 1, \ldots, n$, is compared to μ_0; the numbers of observations greater and less than μ_0 are recorded for each sample. Thus, denote by n_i^+ (n_i^-) the number of observations x_{ij} greater (less) than μ_0 in sample i. Then, the statistic based on signals, SN_i, can be written as follows:

$$SN_i = n_i^+ - n_i^-. \tag{1}$$

It can be shown, based on the continuity of the process, that $SN_i = 2T_i - n$, where $T_i = n_i^+$; furthermore, $T_i \sim \text{Binomial}(n, p)$, with $p = 0.5$, assuming that the median of X_i is μ_0 (i.e., the process is under control at time i). Thus, the control limits and the center line of a Shewhart chart based on signals are given by:

$$\begin{aligned} \text{UCL} &= a, \\ \text{CL} &= 0, \\ \text{LCL} &= -a, \end{aligned}$$

where a is a positive integer such that $0 < a \leq n$.

It is easy to see that the value of a will depend on the desired False Alarm Rate (FAR). Assuming the desired chart is bilateral (or two-sided), a will therefore be such that $\mathbb{P}_{\mu_0|\text{IC}}(SN_i \geq a) \leq \text{FAR}/2$ (due to the process' symmetry). The index/subscript $\mu_0|\text{IC}$ indicates that the control chart's constant is found in case K and under control (IC). The quantiles of the Binomial distribution can be used by rewriting the probability argument in terms of T_i. The methodology is easily adapted for unilateral (or one-sided) cases.

In the proposed package, this technique is implemented under the name *shewhart_sn* and considers the following arguments:

- `X` - a matrix or data frame that should contain the samples for constructing the chart;
- `med` - the median of the process (known or specified), with default 0;
- `far` - the false alarm rate; the default is 0.0027, which generally corresponds to an Average Run Length (ARL) of approximately 370. Depending on the sample size, the desired `far` may be impossible to achieve (due to the discrete behavior of the statistic). In this case, the argument is internally recalculated based on the sample;
- `group_by_col` - a dichotomous argument. If False, rows in `X` are interpreted as subgroups, while columns are observations. If True, columns are interpreted as subgroups;
- `side` - indicates whether the control chart will be bilateral ("two.sided", the default), right unilateral ("upper", adapted from the hypothesis $\mu > \mu_0$), or left unilateral ("lower", adapted from the hypothesis $\mu < \mu_0$).

When called, the function generates a plot with the Shewhart chart based on signs and returns the ARL_0 - the inverse of the specified `far` (or, when not possible, the inverse of the recalculated one).

Most of these arguments reappear in the various control charts implemented in the `npSPC` package. Therefore, only the differences between the methods described from here on will be elucidated.

Shewhart based on signed-ranks

An alternative approach to the previous control chart is using sign statistics combined with a ranking of observations. This technique offers greater statistical efficiency, albeit requiring an additional assumption of symmetry in the distribution of the analyzed data.

Let us consider the same situation as before, but with samples consisting of $n > 1$ random observations collected at regular time intervals from a process with a continuous and symmetric distribution. We assume that the value of μ under control is μ_0, the mean or median of the data.

Let R_{ij} be the rank of the absolute difference $|x_{ij} - \mu_0|$, for $j = 1, \ldots, n$. We define the statistic T_n^+ as the sum of the ranks of the absolute values of deviations corresponding to positive differences, also known as the Wilcoxon signed-rank statistic, which is defined by the following equation:

$$T_n^+ = \sum_{j=1}^{n} \Psi(x_{ij} - \mu_0) R_{ij},$$

where $\Psi(\cdot)$ is an indicator function that returns 1 if the difference $x_{ij} - \mu_0$ is positive, and 0 otherwise. From it, we can obtain the statistic

based on ranked signs, SR_i, which can be written from the linear relationship of T_n^+ as follows:

$$SR_i = 2T_n^+ - \frac{n(n+1)}{2}. \quad (2)$$

In this case, in addition to the ranks corresponding to positive differences, the ranks corresponding to negative differences are also considered.

The exact distribution of the statistic SR_i can be found/calculated by considering all possible permutations of the ranks for a fixed value of n. However, for large values of n, computing all permutations becomes computationally burdensome, and it may be necessary to approximate the distribution to a Normal distribution. In this case, the statistic SR_i follows an approximately Normal distribution with mean $\mathbb{E}[SR_i] = \mathbb{E}[2T_n^+ - n(n+1)/2] = 0$ and variance $\mathbb{V}\text{ar}[SR_i] = n(n+1)(2n+1)/6$ [66].

From the exact or approximate distribution, we can calculate the control limits and center line of a Shewhart chart based on signed-rank as follows:

$$\begin{aligned} \text{UCL} &= a, \\ \text{CL} &= 0, \\ \text{LCL} &= -a, \end{aligned}$$

where the value of a is a positive integer chosen such that $0 < a \leq n(n+1)/2$. Similarly to the case of SN_i, the value of a depends on the desired FAR and is calculated from $\mathbb{P}_{\mu_0|\text{IC}}(SR_i \geq a) \leq \text{FAR}/2$.

The *shewhart_sr* function of the proposed package implements this technique and accepts the same arguments as the *shewhart_sn* function: `X`, `med`, `far`, `group_by_col`, and `side`, including a new argument:

- `exact` - a dichotomous argument. If False, the calculation of the limits will be based on the asymptotic distribution of the statistic. If True, the exact distribution will be used.
  Caution! Using the exact distribution may require a large amount of memory if n exceeds 20; this can cause crashes and even R shutdown.

When using this function, a plot with the Shewhart chart based on signed-rank is generated, and the ARL_0 is returned as a result.

A.1.2. CuSum control charts

CuSum based on signs

Suppose it is desired to identify whether $\mu > \mu_0$, where μ_0, as before, represents the median, known or specified, of the in-control process. The statistic for the CuSum chart based on signs is given by:

$$SN_i^+ = \max\left(0, SN_{i-1}^+ + SN_i - k\right), \quad \text{for } i = 1, 2, \ldots, \quad (3)$$

with initial value $SN_0^+ = 0$. The technique will indicate an out-of-control condition when $SN_i^+ \geq h$.

Similarly, if the interest is in identifying whether $\mu < \mu_0$, then the following statistic is used:

$$SN_i^- = \min\left(0, SN_{i-1}^- + SN_i + k\right), \quad \text{for } i = 1, 2, \ldots, \quad (4)$$

with initial value $SN_0^- = 0$, and an out-of-control condition indicated when $SN_i^- \leq -h$.

In the bilateral case, both statistics are considered, and the process will be deemed out of control when $SN_i^+ \geq h$ or $SN_i^- \leq -h$.

Notably, $h > 0$ and $k > 0$ affect the performance of the control chart and must be specified to achieve the desired FAR when the process is in control and, at the same time, the smallest possible ARL_1 otherwise (i.e., when the process is out of control). Future versions of the package should provide functions to indicate the best choice for both values; for now, they must be provided by the user.

The function implementing this technique in the package was named *cusum_sn* and, in addition to `X`, `med`, `group_by_col`, and `side`, already described previously, it has the following arguments:

- `k` - reference or slack value, similar to the parametric case. The default is 0.5;
- `h` - defines the control chart limits. UCL will be given by h, while LCL will be $-h$. Default is 5;
- `plot` - if True, the function will generate the control chart plot. Otherwise, it will return the index of the first sample where the process went out of control.

It is important to note that the default values of the functions may not perform well depending on the data set (the resulting FAR depends on the sample size used), unlike what was implemented in the *shewhart_sn* function, where the homonymous parameter controlled the FAR.

CuSum based on signed-ranks

The only difference between this control chart and the previous one is the substitution of the sign statistic SN_i in Eqs. (3) and (4) by the ranked sign statistic SR_i, defined in Eq. (2). Therefore, if the interest is in identifying whether $\mu > \mu_0$, the process will be out of control if $SR_i^+ \geq h$. If the interest is in identifying whether $\mu < \mu_0$, the process will be out of control if $SR_i^- \leq -h$. For the bilateral control chart, an alarm will occur if $SR_i^+ \geq h$ or $SR_i^- \leq -h$.

This technique can be called through the *cusum_sr* function, whose arguments are the same as the *cusum_sn* function. However, it is worth noting that defining the same values for all parameters will generally not result in equivalent FAR and ARL_1 for both charts. Since the base statistics are different, the values of h and k will influence the performance of each type of chart differently.

A.1.3. EWMA control charts

EWMA based on signs

Consider again the statistic SN_i, defined in Eq. (1). For the bilateral case, the statistic used in the EWMA chart based on signs is defined as follows:

$$Z_i = \lambda SN_i + (1-\lambda)\, Z_{i-1}, \quad \text{for } i = 1, 2, \ldots,$$

with initial value $Z_0 = 0$ and $0 < \lambda \leq 1$. The center line and exact control limits are given by:

$$\begin{aligned} \text{UCL} &= L\sqrt{\frac{\lambda n}{2-\lambda}\left(1-(1-\lambda)^{2i}\right)}, \\ \text{CL} &= 0, \\ \text{LCL} &= -L\sqrt{\frac{\lambda n}{2-\lambda}\left(1-(1-\lambda)^{2i}\right)}. \end{aligned}$$

The EWMA chart based on a right-sided hypothesis test has the statistic given by $Z_i^+ = \max\left(0, Z_i\right)$, while the EWMA chart based on a left-sided hypothesis test uses the statistic $Z_i^- = \min\left(0, Z_i\right)$.

Similar to the CuSum case, in the *ewma_sn* function, the FAR is not directly passed as an argument. Here, the control limits, ARL_1, and FAR are controlled through two parameters: λ and L; where the former is a constant called the smoothing parameter and the latter is a graphical constant, which will affect the control limits. The other arguments of the function are as in *shewhart_sn*, with the addition of the `plot` parameter, already defined previously.

EWMA based on signed-ranks

The most important differences between the functions *ewma_sn* and *ewma_sr*, besides the assumption of symmetry in the data distribution in the latter, are:

- Use of the statistic SR_i, defined in Eq. (2), instead of the statistic SN_i;
- Substitution of n in the control limits of the statistic by $\sigma^2_{SR_i}$.

Thus,

$$Z_i = \lambda SR_i + (1-\lambda)\, Z_{i-1}, \quad \text{for } i = 1, 2, \ldots,$$

with the center line and control limits given by:

$$\mathrm{UCL} = L\sqrt{\frac{\lambda\sigma^2 SR_i}{2-\lambda}\left(1-(1-\lambda)^{2i}\right)},$$

$$\mathrm{CL} = 0,$$

$$\mathrm{LCL} = -L\sqrt{\frac{\lambda\sigma^2 SR_i}{2-\lambda}\left(1-(1-\lambda)^{2i}\right)},$$

where $\sigma^2_{SR_i} = n(n+1)(2n+1)/6$ is the variance of SR_i, and λ and L are as before.

The arguments of *ewma_sr* are the same as *ewma_sn*.

A.2. Case U (unknown in-control limits)

The main difference between the charts of cases U and K lies in the fact that, in case U, there is a need to estimate the parameter of interest. Such estimation and the control chart construction are typically performed using a phase 1 sample; thus, subsequently, phase 2 samples from the same process can be monitored.

A.2.1. Shewhart control charts

Shewhart based on precedence statistic

According to Gibbons and Chakraborti [67], the precedence test is a popular nonparametric test for two samples. It can be seen as a generalization of the sign test for one sample. Adapting this methodology to Shewhart control charts assumes that the data distribution, in addition to being continuous, is of location. Thus, suppose a phase 1 sample with m observations, $X_1, \ldots, X_m$, has a cumulative distribution function (CDF) $F(x) = H(x-\theta_1)$, with θ_1 being the location parameter of interest. The m observations are ordered (ascending), and two order statistics $X_{(a:m)}$ and $X_{(b:m)}$, for data a and b, with $1 \le a < b \le m$, are defined according to some control chart performance criterion.

These order statistics will be the control limits for the Shewhart chart based on the precedence statistic, that is,

$$\widehat{\mathrm{LCL}} = X_{(a:m)} \quad \text{and} \quad \widehat{\mathrm{UCL}} = X_{(b:m)}.$$

In phase 2, test samples independent of each other and of the phase 1 sample are randomly taken; it is assumed that their respective CDF is given by $G_i(y) = H(y-\theta_i)$. The interest is, in the bilateral case, to test if $\theta_i \neq \theta_1$, for some $i = 2, 3, \ldots$ (the process can be easily adapted if the interest is to test if $\theta_i > \theta_1$ or if $\theta_i < \theta_1$). Thus, assume that the test samples have size n and that the ith sample has observations denoted by $\boldsymbol{Y}_i = (Y_{i1}, \ldots, Y_{in})$. Then, the chart statistic will be $Y_{(j:n)}$, the jth order statistic, for a given value j, with $j = 1, 2, \ldots, n$. Typically and by default in the *shewhart_prec* function, j will be such that $Y_{(j:n)}$ is the median of $\boldsymbol{Y}_i$ (this formulation is also referred to as the Med chart, with Med being an abbreviation for median).

That said, it is easy to see that the process will result in an alarm if $Y_{(j:n)} < X_{(a:m)}$ or if $Y_{(j:n)} > X_{(b:m)}$. On the other hand, if we let W_j, $j = 1, 2, \ldots, n$, denote the number of observations in phase 1 that are smaller than $Y_{(j:n)}$, we can redefine the control limits of the chart. That is, the process will be out of control if $W_j > b-1$ or $W_j < a$. This change is convenient because W_j has an exact and relatively simple distribution to calculate (Eq. (5)), allowing the FAR to be also defined precisely and directly.

$$\mathbb{P}_{\mathrm{IC}}\left(W_j = w\right) = \frac{\binom{w+j-1}{w}\binom{m+n-j-w}{m-w}}{\binom{m+n}{m}}, \tag{5}$$

with $w = 0, 1, \ldots, m$, where the index/subscript IC indicates that the probability distribution of W_j is in control.

Note that to define the control limits with a FAR of, say, 0.0027, it suffices to find a and b such that $\mathbb{P}_{\mathrm{IC}}(W_j < a) = 0.00135$ and $\mathbb{P}_{\mathrm{IC}}(W_j \ge b) = 0.00135$.

The arguments of the *shewhart_prec* function are defined below:

- `X` - a matrix or vector containing phase 1 data;
- `Y` - a matrix containing test data;
- `j` - an integer between 1 and the number of rows of Y; by default, it is the largest integer less than half the number of rows of Y (in this case, $Y_{(j:n)}$ will be the median for each $\boldsymbol{Y}_i$);
- `far` - the false alarm rate; the default is 0.0027, which generally corresponds to an ARL_0 of approximately 370. Depending on the sample size, the desired FAR may be impossible to obtain (due to the discrete behavior of the statistic W_j). In this case, the argument is internally recalculated based on the sample;
- `group_by_col` - a dichotomous argument. If False, the rows in X are interpreted as subgroups, while the columns as observations. If True, the columns are interpreted as subgroups;
- `side` - indicates whether the control chart will be bilateral ("two.sided", the default), right unilateral ("upper", adapted from the hypothesis $\mu > \mu_0$), or left unilateral ("lower", adapted from the hypothesis $\mu < \mu_0$).

Similar to the *shewhart_sn* function, when called, the *shewhart_prec* function generates a control chart based on the precedence statistic. It returns the ARL_0 - the inverse of the specified FAR (or, when not possible, the inverse of the recalculated FAR).

Shewhart based on Mann–Whitney statistic

Making $X_1, \ldots, X_m$ and $Y_{i1}, \ldots, Y_{in}$ as in the previous section, for each $\boldsymbol{Y}_i$, the Mann–Whitney statistic is defined as follows:

$$M_{iXY} = \sum_{k=1}^{m}\sum_{j=1}^{n} I(Y_{ij} > X_k), \tag{6}$$

where $i = 1, 2, \ldots$. Note that M_{iXY} varies between 0 and mn, and large (small) values of M_{iXY} indicate a positive (negative) shift in location. The control chart will result in an alarm if $M_{iXY} > b_{mn}$ or $M_{iXY} < a_{mn}$, where a_{mn} and b_{mn} are the control limits, defined according to some chart performance criterion (i.e., to achieve the desired FAR). For the bilateral case, it is worth noting the relationship $a_{mn} = mn - b_{mn}$; extending the technique to unilateral cases is straightforward, with the caveat that, for the same FAR, the a_{mn} (b_{mn}) for the bilateral case will generally be different from the a_{mn} (b_{mn}) for the left (right) unilateral case.

The arguments of *shewhart_mw* are similar to those of *shewhart_prec*. However, the former function does not include the argument `far`, which should be controlled through `cl`, representing b_{mn} when `side = ''two.sided''` or `side = ''upper''`, and a_{mn} otherwise. Another difference is this function's return of the `plot` argument.

A.2.2. CuSum control charts

CuSum based on exceedance statistic

As before, consider a phase 1 sample, $X_1, \ldots, X_m$ ($m > 1$), with CDF $F(x)$. Let $Y_{i1}, \ldots, Y_{in}$, with $i = 1, 2, \ldots$, be the ith test sample of size n with CDF $G(y)$. Both F and G are continuous and unknown distribution functions.

Now, let $U_{i,r}$ be the number of observations in the ith sample of $\boldsymbol{Y}$ that exceed $X_{(r)}$. $U_{i,r}$ is called the exceedance statistic, and therefore, the CuSum chart constructed from it has statistics given by:

$$U_i^+ = \max\left(0, U_{i-1}^+ + (U_{i,r} - nd) - k\right), \quad \text{for } i = 1, 2, \ldots,$$

with initial value $U_0^+ = 0$. The technique will indicate an out-of-control state when $U_i^+ \ge h$; and

$$U_i^- = \min\left(0, U_{i-1}^- + (U_{i,r} - nd) + k\right), \quad \text{for } i = 1, 2, \ldots,$$

with initial value $U_0^- = 0$. The technique will indicate an out-of-control state when $U_i^- \le -h$.

Analogously to the previously defined CuSums, in the bilateral case, both statistics are considered, and the process will be considered out of control when $U_i^- \leq -h$ or $U_i^+ \geq h$.

Here, h and k are as before; $X_{(r)}$ is the rth observation in the phase 1 sample, ordered in ascending order, and $d = (m - r + 1)/(m + 1)$.

Thus, the only argument different from the *cusum_ex* function is `r`, which by default is set as the number of elements in the `X` vector divided by 2. The idea behind this argument is similar to j in the *shewhart_mw* function; the difference is that j is expressed in terms of the number of observations from one of the phase 2 samples, whereas r is expressed in terms of the phase 1 sample.

CuSum based on Wilcoxon rank-sum statistic

The statistics for the CuSum chart based on the Wilcoxon rank-sum statistic are given by:

$$W_i^+ = \max\left(0, W_{i-1}^+ + \left(W_i - \frac{n(m+n+1)}{2}\right) - k\right),$$

for $i = 1, 2, \ldots$, with initial value $W_0^+ = 0$. The technique will indicate an out-of-control state when $W_i^+ \geq h$; and

$$W_i^- = \min\left(0, W_{i-1}^- + \left(W_i - \frac{n(m+n+1)}{2}\right) + k\right),$$

for $i = 1, 2, \ldots$, with initial value $W_0^- = 0$. The technique will indicate an out-of-control state when $W_i^- \leq -h$. Here, W_i indicates the Wilcoxon ranked sum statistic, which can be written using the Mann–Whitney statistic, defined in Eq. (6); thus, $W_{XY} = M_{XY} + m(m+1)/2$.

The arguments of *cusum_wr* are the same as *cusum_ex*, except for `r`, which does not exist in the former function. Just like in case K, the values of h and k affect the performance of both charts differently.

A.2.3. EWMA control charts

EWMA based on exceedance statistic

Again, consider the exceedance statistic $U_{i,r}$. For the bilateral case, the statistic used in the EWMA chart is defined as follows:

$$Z_i = \lambda U_{i,r} + (1 - \lambda) Z_{i-1}, \quad \text{for } i = 1, 2, \ldots,$$

with initial value $Z_0 = 0$ and $0 < \lambda \leq 1$. The center line and exact control limits are given by:

$$\mathrm{CL} = n(1-a)\left(1 - (1-\lambda)^i\right),$$

$$\mathrm{UCL} = \mathrm{CL} + L\sqrt{\left(\frac{na(1-a)}{m+2}\right)\left(n\left(1-(1-\lambda)^i\right)^2 + \frac{\lambda(m+1)}{2-\lambda}\left(1-(1-\lambda)^{2i}\right)\right)},$$

$$\mathrm{LCL} = \mathrm{CL} - L\sqrt{\left(\frac{na(1-a)}{m+2}\right)\left(n\left(1-(1-\lambda)^i\right)^2 + \frac{\lambda(m+1)}{2-\lambda}\left(1-(1-\lambda)^{2i}\right)\right)},$$

with $a = r/(m+1)$.

The adaptation for unilateral charts is analogous to the EWMA case K.

All parameters used in the function *ewma_ex* have been presented previously: `X`, `Y`, `lambda`, `L`, `r`, `group_by_col`, `plot`, and `side`.

EWMA based on Wilcoxon rank-sum statistic

The main difference between the functions *ewma_ex* and *ewma_wr* lies in the use of the statistic W_i instead of $U_{i,r}$, and in the substitution of n in the control limits of the statistic by σ_W^2.

Therefore,

$$Z_i = \lambda W_i + (1 - \lambda) Z_{i-1}, \quad \text{for } i = 1, 2, \ldots$$

and $0 < \lambda \leq 1$, with the center line and control limits given by:

$$\mathrm{CL} = \frac{n(m+n+1)}{2},$$

$$\mathrm{UCL} = \mathrm{CL} + L\sqrt{\frac{\lambda\sigma_W^2}{2-\lambda}\left(1-(1-\lambda)^{2i}\right)},$$

$$\mathrm{LCL} = \mathrm{CL} - L\sqrt{\frac{\lambda\sigma_W^2}{2-\lambda}\left(1-(1-\lambda)^{2i}\right)},$$

in which $\sigma_W^2 = mn(m+n+1)/12$ represents the variance of W_i, and L and λ remain as before.

The arguments for *ewma_wr* are the same as those for *ewma_ex*, except for `r`, which does not exist in the former function.

Data availability

The data sets and R codes can be found at: https://github.com/CER-UFBA/npSPC.

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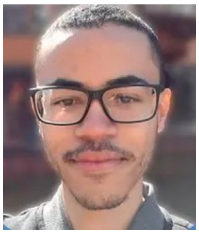
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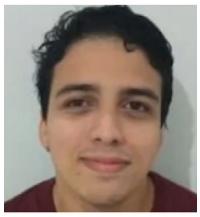
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